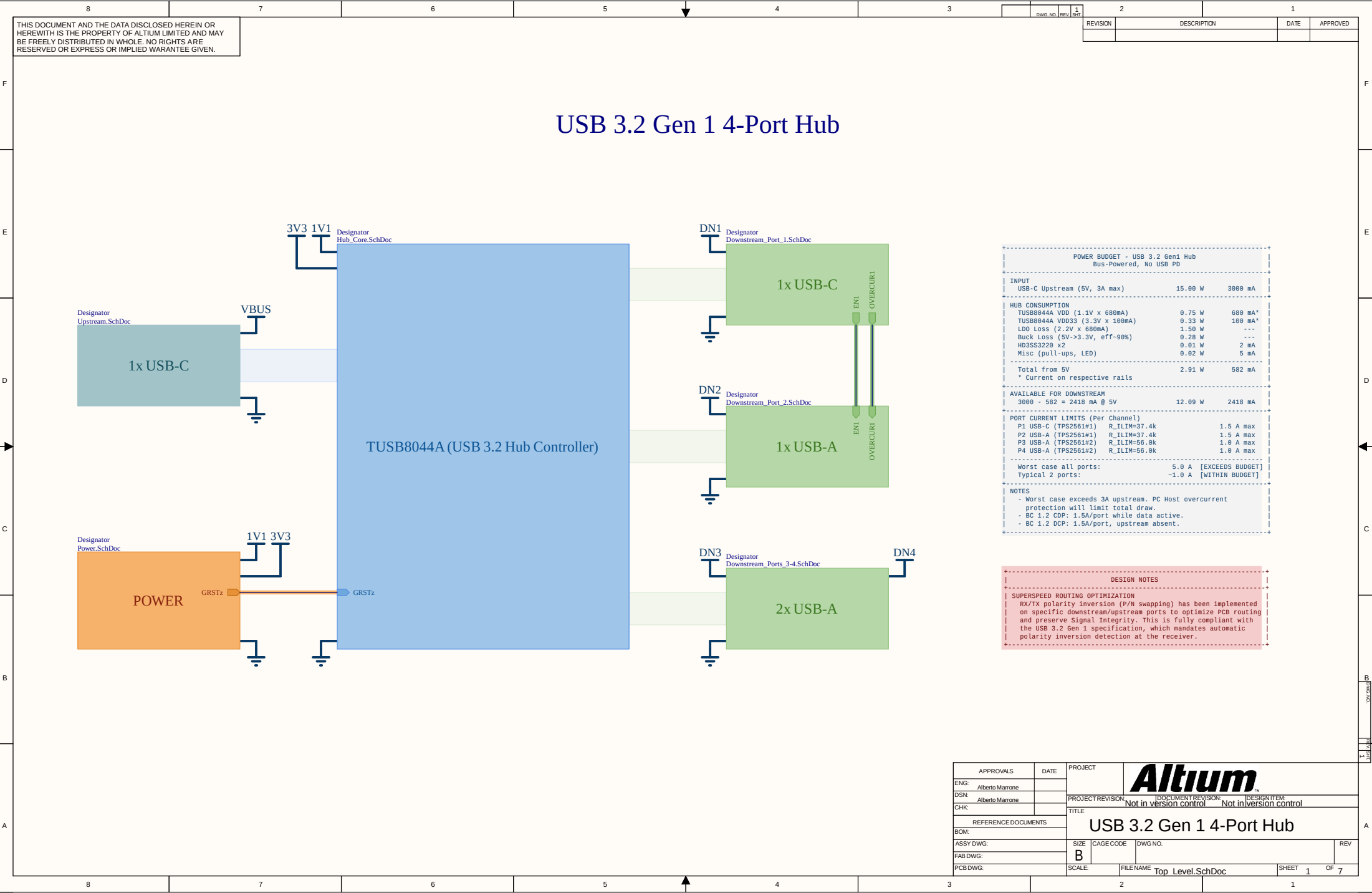




2		1	
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ESD Protection:

- PUSB3FR42: 0.29pF, ± 15 kV, USB3 protection
- TPD4E05U06: 0.5pF, USB2/CC protection
- SMAJ5.0A: VBUS TVS, clamps surges to ground

NOTE: SuperSpeed P/N Polarity Swapped for PCB routing optimization.

[illegible]

VBUS_DET divider:
 $V_{pin} = 5V \times 95k / (909k + 95k) = 0.47V$
 Within HD3SS3220 detect range

Pin 48 connection diagram for TUSB0044ARPCR. The diagram shows a yellow box labeled 'U3A' containing pins 48 through 59. Pin 48 is labeled 'USB_VBUS'. Pins 49 and 50 are labeled 'USB_DP' and 'USB_DM' respectively. Pins 51 and 52 are labeled 'USB_SSTXP_UP' and 'USB_SSTXM_UP' respectively. Pins 53 and 54 are labeled 'USB_SSRXP_UP' and 'USB_SSRXM_UP' respectively. A red dashed box highlights pins 51, 52, 53, and 54. A red circle with a crosshair is placed below the dashed box, labeled 'USB SS'. A red arrow points from the 'USB SS' label to the dashed box.

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ENG: Alberto Marrone			PROJECT REVISION: Not in version control			DOCUMENT REVISION: Not in version control
DSN: Alberto Marrone			DESIGN ITEM: Not in version control			
CHK:			TITLE			
REFERENCE DOCUMENTS			USB 3.2 Gen 1 4-Port Hub			
BOM:						
ASSY DWG:			SIZE	CAGE CODE	DWG NO.	REV
FAB DWG:			B			
PCB DWG:			SCALE:	FILE NAME	Upstream_SchDoc	SHEET 3 OF 7

[illegible]

Circuit diagram of a crystal oscillator. The crystal X1 (24MHz) is connected between XTAL_IN and XTAL_OUT. XTAL_OUT is connected to XTAL_IN through a resistor R20 (1M). Both XTAL_IN and XTAL_OUT are connected to ground through capacitors C37 and C38 (15pF).

$$C1/C2 = (C1-5(\text{stray})) * 2 = 10-15\text{pF}$$

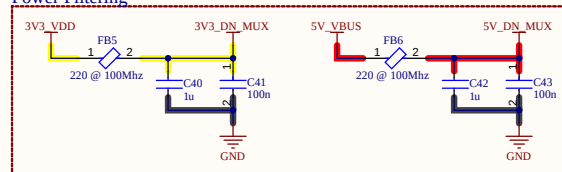
EEPROM WP CONFIGURATION:
 If R_PD placed -> Write enabled
 If R_PU placed -> Write disabled

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REFERENCE DOCUMENTS			PROJECT REVISION:	DOCUMENT REVISION:	DESIGN ITEM:
BOM:			TITLE		
			USB 3.2 Gen 1 4-Port Hub		
ASSY DWG:	SIZE	CAGE CODE	DWG NO.		REV
FAB DWG:	B				
PCB DWG:	SCALE:	FILENAME	Hub_Core_SchDoc		SHEET 4 OF 7

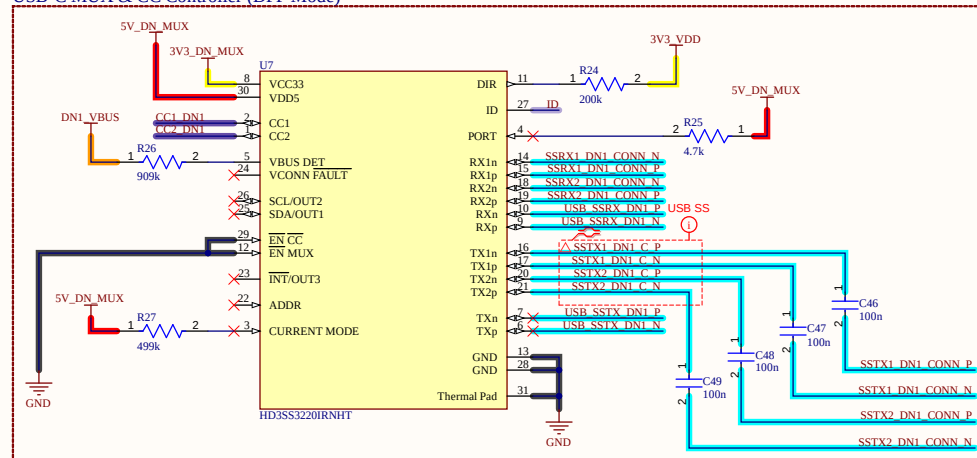
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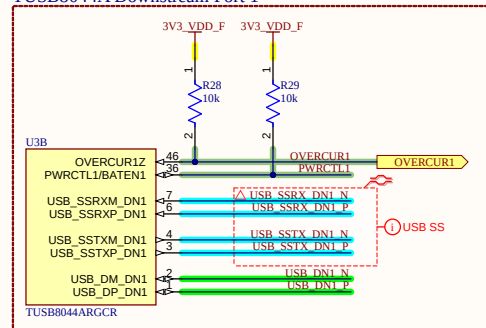
Power Filtering



USB-C MUX & CC Controller (DFP Mode)

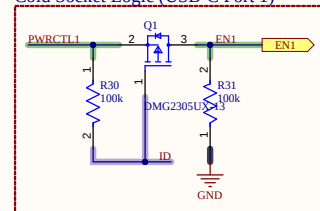


TUSB8044A Downstream Port 1

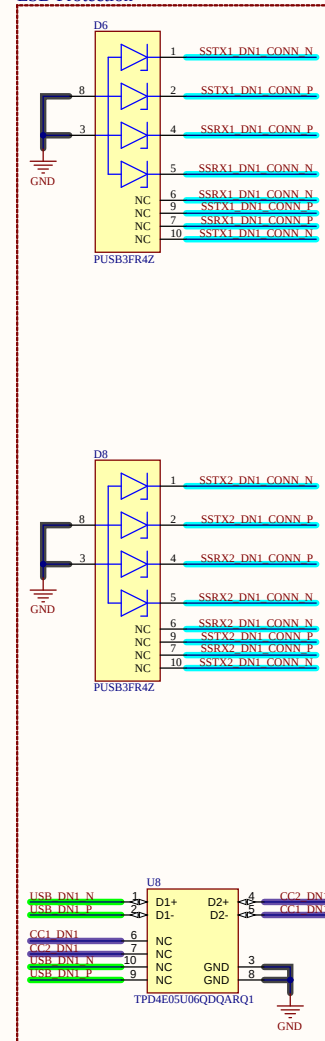


NOTE: SuperSpeed P/N Polarity Swapped for PCB routing optimization.

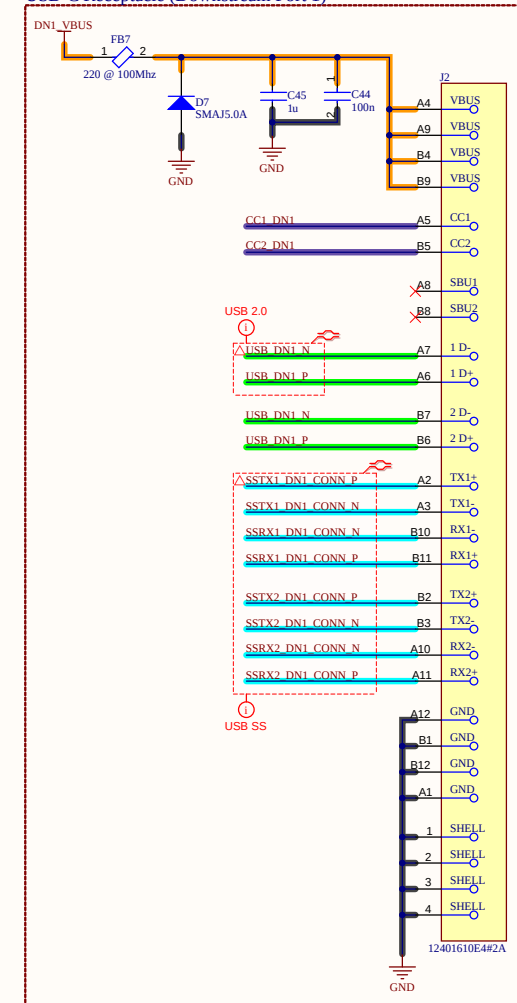
Cold Socket Logic (USB-C Port 1)



ESD Protection



USB-C Receptacle (Downstream Port 1)



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CHK:			
BOM:			
ASSY DWG:			
FAB DWG:			
PCB DWG:			

3V3_VDD_F

3V3_VDD_F

1

2

R37
10k

R38
10k

1

2

U3C

OVERCUR2Z
PWRCRTL2/BATEN2

47

35

34

34

11

12

10

10

OVERCUR2
PWRCRTL2

47

35

34

34

11

12

10

10

USB_SSRX_DN2 N

USB_SSRX_DN2 P

USB_SSTX_DN2 N

USB_SSTX_DN2 P

USB_DM_DN2

USB_DP_DN2

USB SS

USB D2.0

TUSB8044ARGCR

The diagram shows the USB3FR4Z component with the following pin connections:

- Pin 1:** USB_SSRX_DN2_N (Crossed out with a red X)
- Pin 2:** USB_SSRX_DN2_P (Crossed out with a red X)
- Pin 3:** GND (Connected to a ground symbol)
- Pin 4:** USB_SSTX_DN2_N
- Pin 5:** USB_SSTX_DN2_P
- Pin 6:** USB_SSTX_DN2_P
- Pin 7:** USB_SSRX_DN2_P
- Pin 8:** USB_SSTX_DN2_N
- Pin 9:** USB_SSTX_DN2_N
- Pin 10:** USB_SSRX_DN2_N

NC (Not Connected) is indicated for pins 6, 7, 8, and 9.

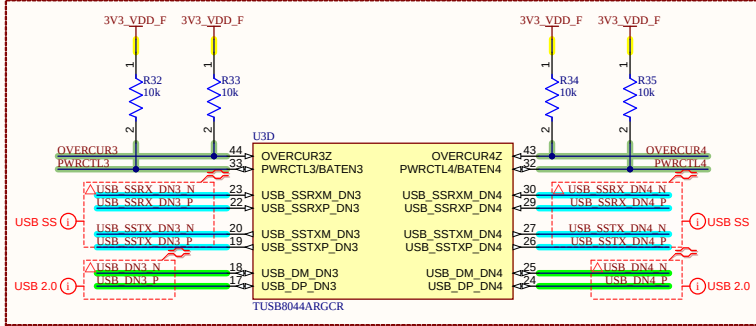
[illegible]

APPROVALS		DATE		PROJECT  PROJECT REVISION DOCUMENT REVISION DESIGN ITEM	
ENG: Alberto Marrone					
DSN: Alberto Marrone					
CHK:					
REFERENCE DOCUMENTS				TITLE	
BOM:				USB 3.2 Gen 1 4-Port Hub	
ASSY DWG:				SIZE	
FAB DWG:				CAGE CODE	
PCB DWG:				DWG NO.	
				REV	
				B	
				SCALE	
				FILE NAME	
				Downstream Port 2 SchDoc	
				SHEET	
				6 OF 7	

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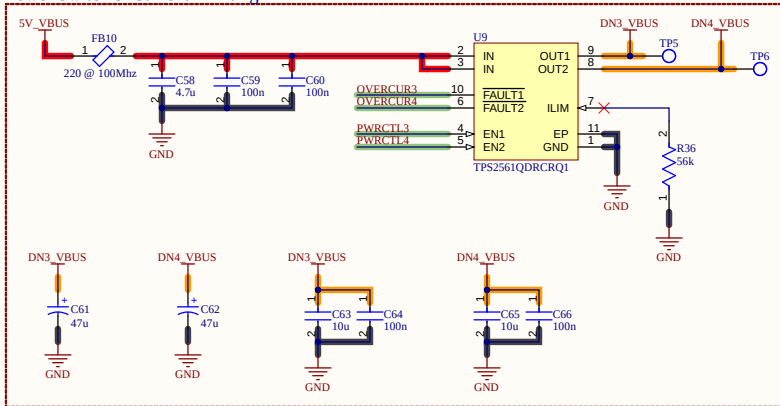
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TUSB8044A Downstream Ports 3-4

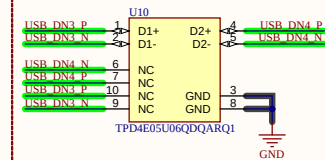
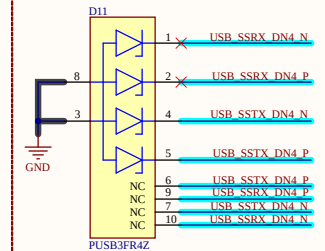
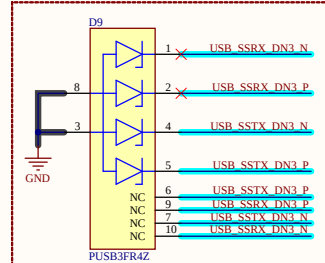


TPS2561 Current Limit Configuration:
 $I_{OS(yp)} = 56000 / R_{ILIM} [k\Omega] (mA)$
 $R_{ILIM} = 37.4k\Omega \rightarrow 1.5A/ch (Ports 1+2)$
 $R_{ILIM} = 56k\Omega \rightarrow 1.0A/ch (Ports 3+4)$
Per-channel limit, NOT shared between channels.
Overcurrent triggers output disconnect + FAULT LOW.
Auto-retry after fault condition clears.

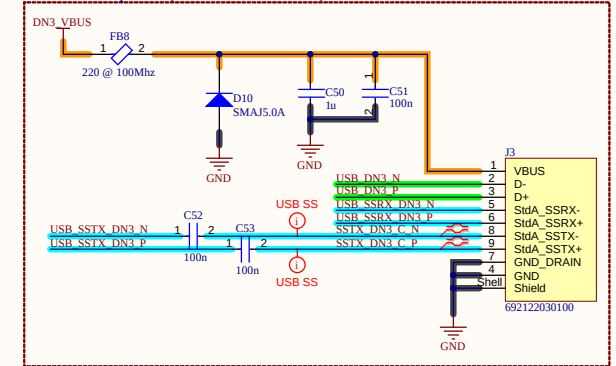
Power Switch & Current Limiting



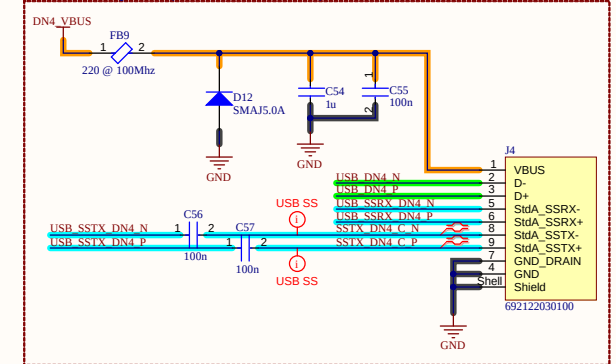
ESD Protection



USB-A Receptacle (Downstream Port 3)



USB-A Receptacle (Downstream Port 4)



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CHK:			
BOM:			
ASSY DWG:			
FAB DWG:			
PCB DWG:			